

Data Visualization

The background features a 3D architectural design. On the right side, a series of horizontal, rectangular blocks are stacked to form a staircase that recedes into the distance. These blocks have a dark, metallic appearance with a blue-to-purple gradient. On the left side, there is a large, dark, triangular shape that points towards the top right, partially overlapping the staircase. The overall color palette is dark and futuristic, with a gradient from black and dark blue to a lighter, hazy blue at the top right.

LEÓN JARAMILLO

SOFTSERVE ACADEMY MENTOR

SoftServe



LEARNING OBJECTIVES

- Explain why visualization matters
- Distinguish exploration vs. presentation
- Choose charts based on data type
- Identify EDA techniques
- Recognize when interactivity helps

Raw data

Visual exploration

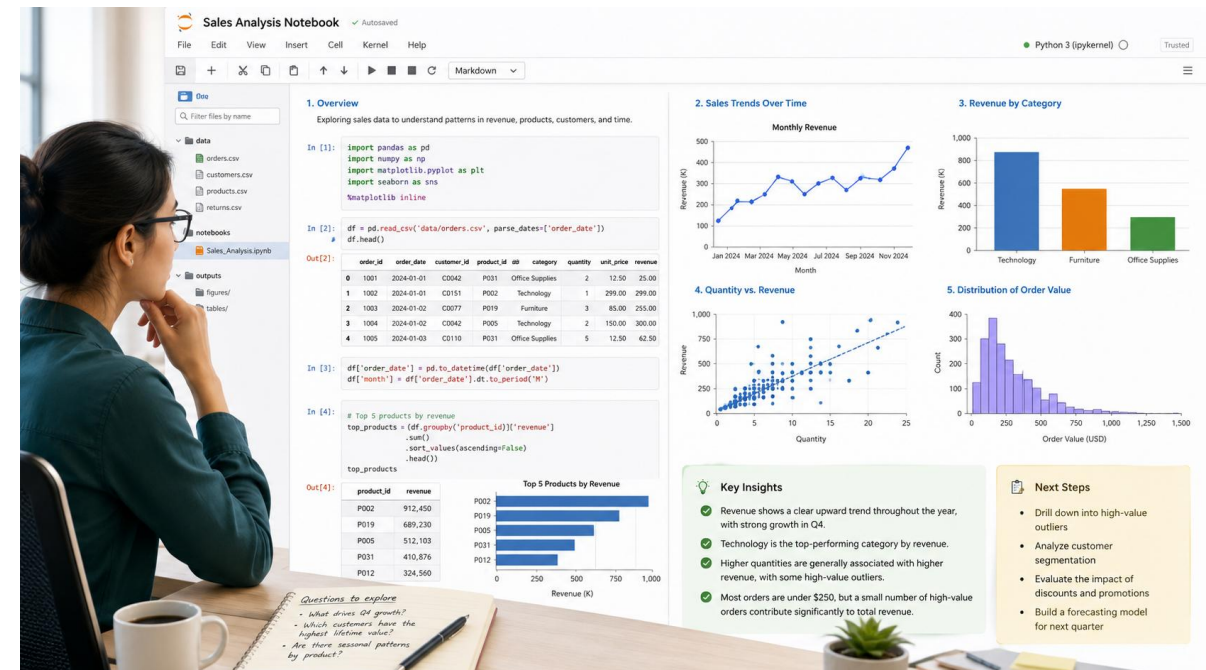
Insight

Decision



DATA VISUALIZATION & EXPLORATION

Visualization helps us understand data before modeling or decision-making.





WHY VISUALIZATION MATTERS

- Raw data is hard to interpret.
- Charts reveal trends, clusters, outliers, and relationships faster than tables.



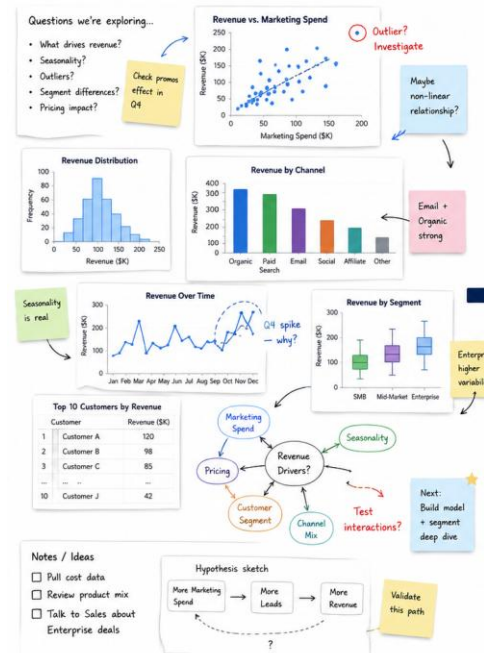


EXPLORATION VS. PRESENTATION

Exploration: Finding what is true.

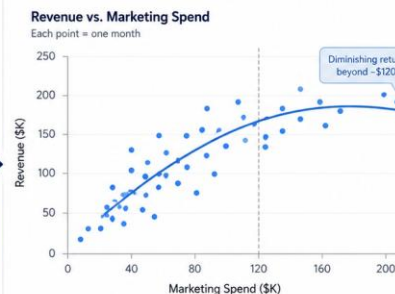
Presentation: Communicating what you found clearly to others.

Explore: messy, many charts



Present: polished, focused story.

Marketing investment drives disproportionate revenue growth.
Revenue increases rapidly with higher marketing spend, with diminishing returns beyond ~\$120K.



Key takeaway
Increasing marketing spend up to ~\$120K is associated with strong revenue gains. Beyond that point, returns plateau.



Recommendation
Increase marketing spend to ~\$120K/month and focus on improving campaign efficiency beyond that level.



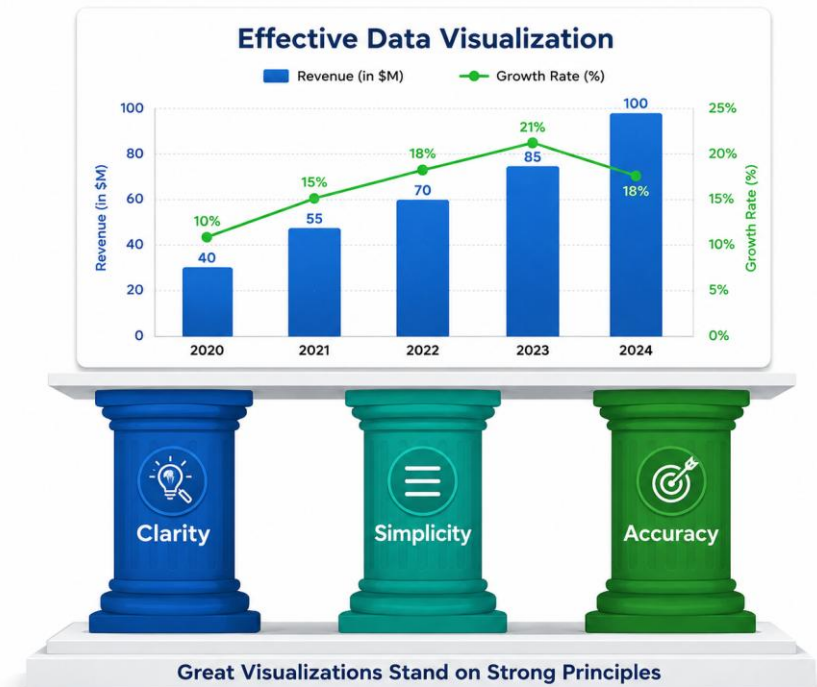
PRINCIPLES OF EFFECTIVE VISUALIZATION

Principles

- Clarity
- Simplicity
- Accuracy

What to avoid

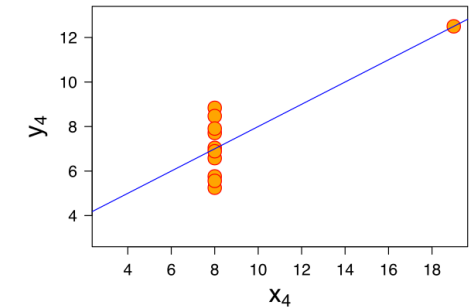
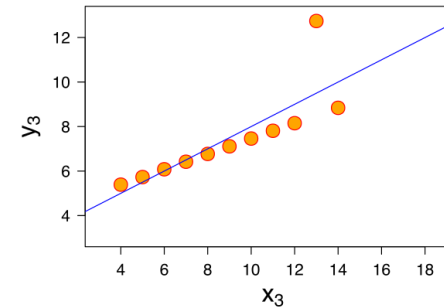
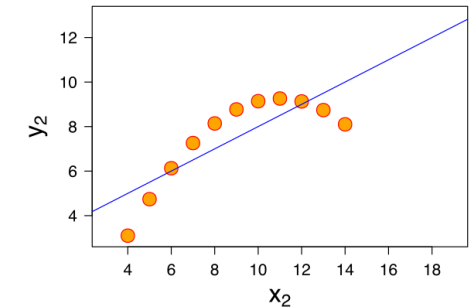
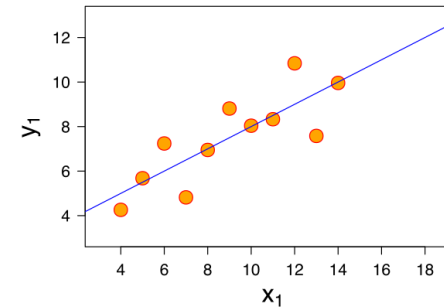
- Misleading scales
- Clutter
- Decorative noise.





VISUALIZATION VS. SUMMARY STATISTICS

- Mean, variance, and correlation can hide very different patterns.
- Always inspect distributions visually.





TYPES OF DATA

- Nominal
- Ordinal
- Interval
- Ratio

The data type determines which visual encodings make sense.

	Categorical	Numerical
Unordered	 Nominal color, country, product type	 Interval temperature (°C), calendar year
Ordered	 Ordinal rating, education level, size (S/M/L)	 Ratio age, income, distance



PYTHON VISUALIZATION TOOLKIT

Matplotlib for core plotting

Seaborn for statistical graphics

Plotly and Bokeh for interactive visuals

matplotlib



seaborn



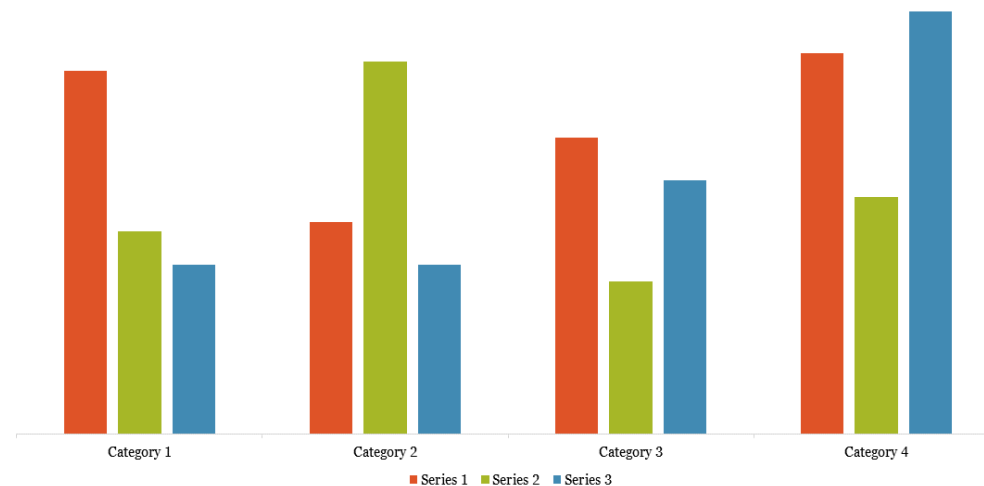
plotly



1D CATEGORICAL DATA: BAR CHARTS

- Use bar charts to compare category counts.
- Avoid connecting category bars with lines.

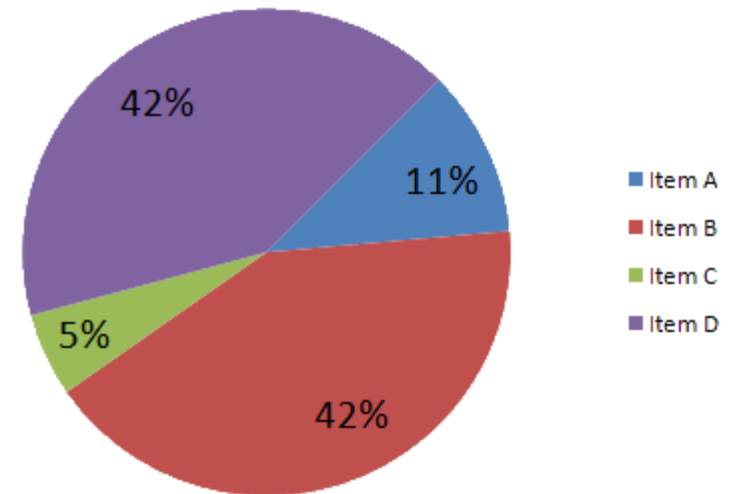
Blank Bar Graph Template





PIE CHARTS: USE CAREFULLY

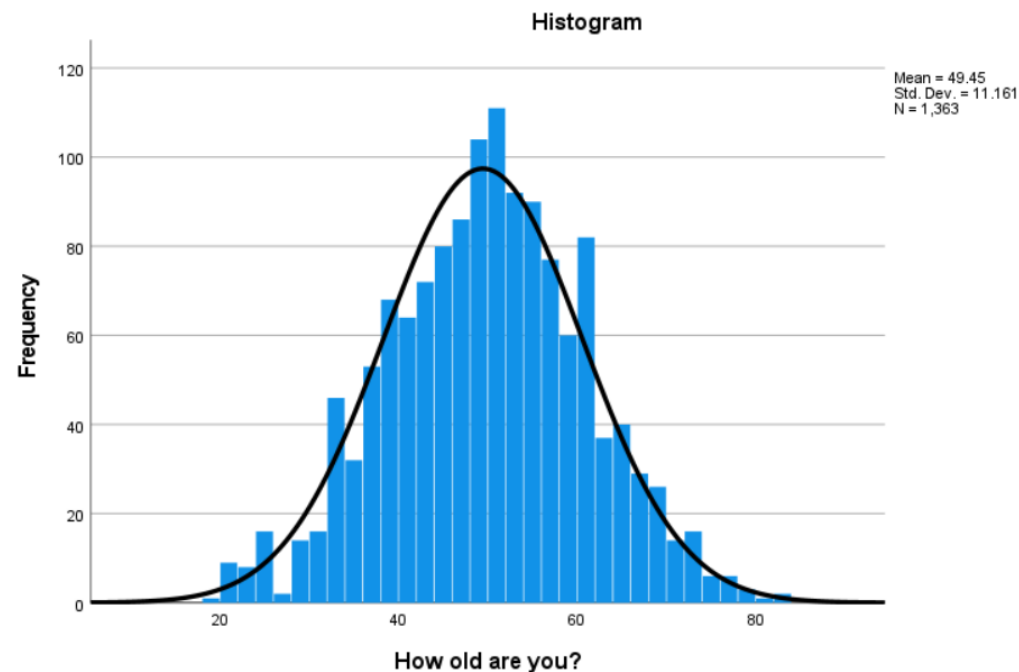
- Pie charts are often hard to compare, especially with many categories.
- Bar charts are usually clearer.





1D NUMERICAL DATA: HISTOGRAMS

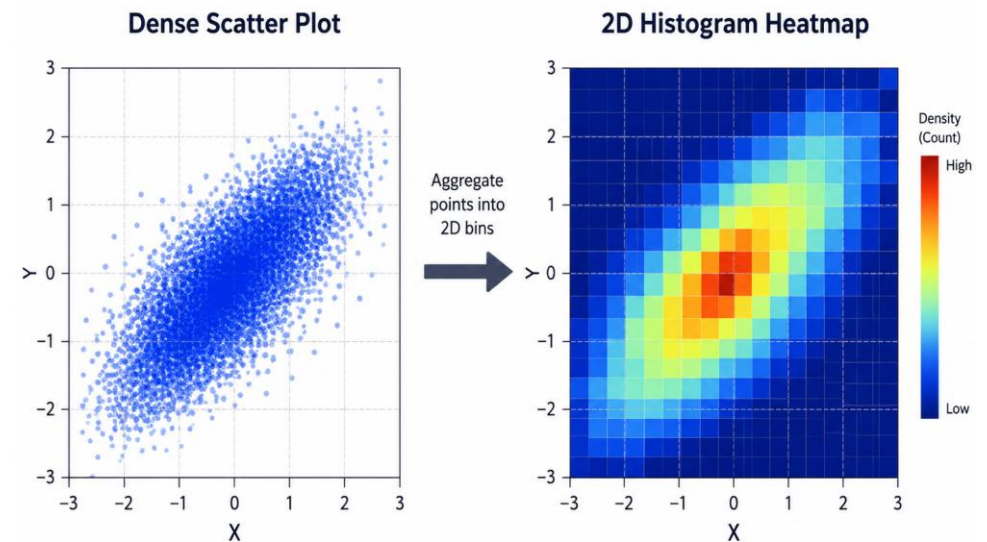
- Histograms show how numerical values are distributed using bins.
- Useful for skew, spread, peaks, and unusual values.





2D NUMERICAL DATA: SCATTER PLOTS

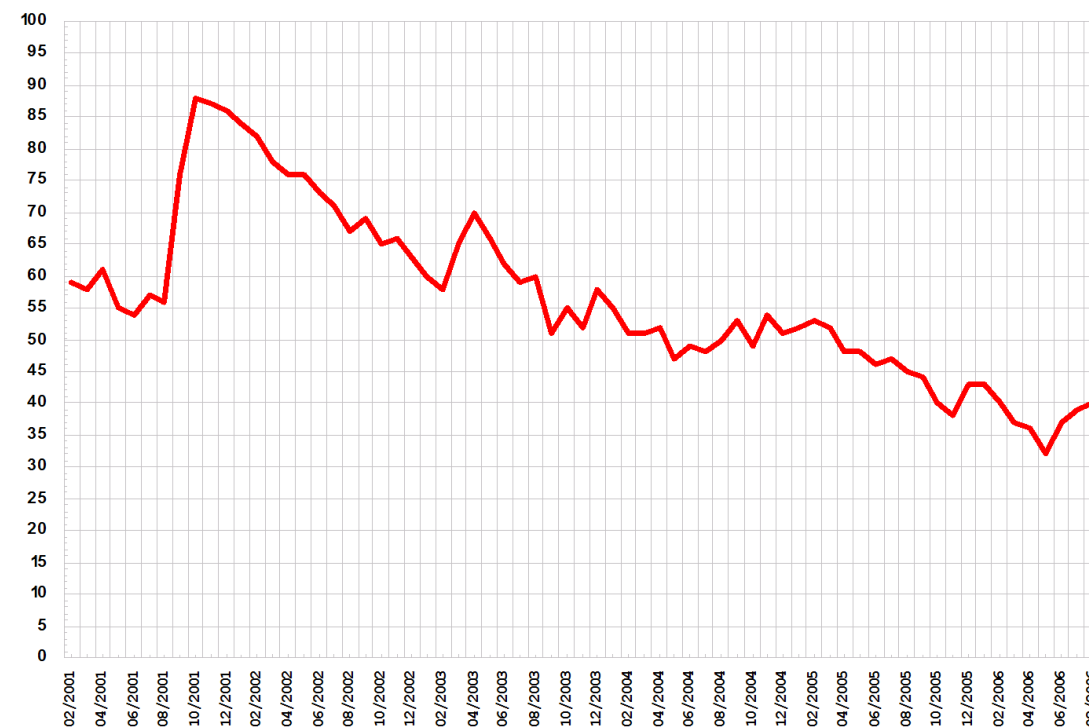
- Scatter plots reveal relationships, clusters, and nonlinear patterns between two numerical variables.
- Too many points can hide structure. Use transparency, smaller points, density plots, or 2D histograms.





SEQUENTIAL NUMERIC DATA: LINE PLOTS

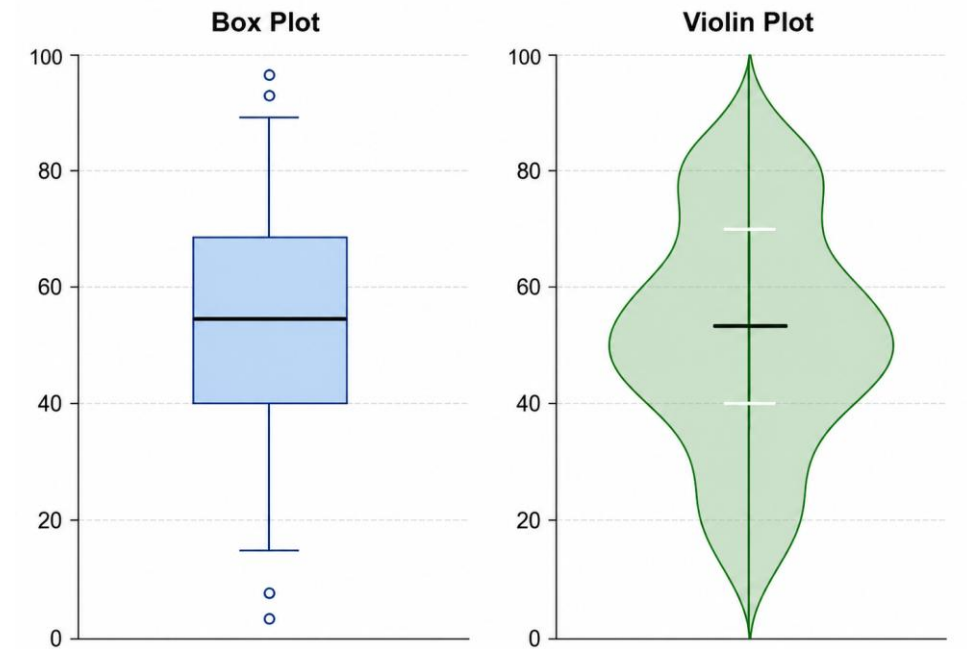
Use line plots when one variable has a meaningful sequence, especially time series.





CATEGORICAL + NUMERICAL DATA

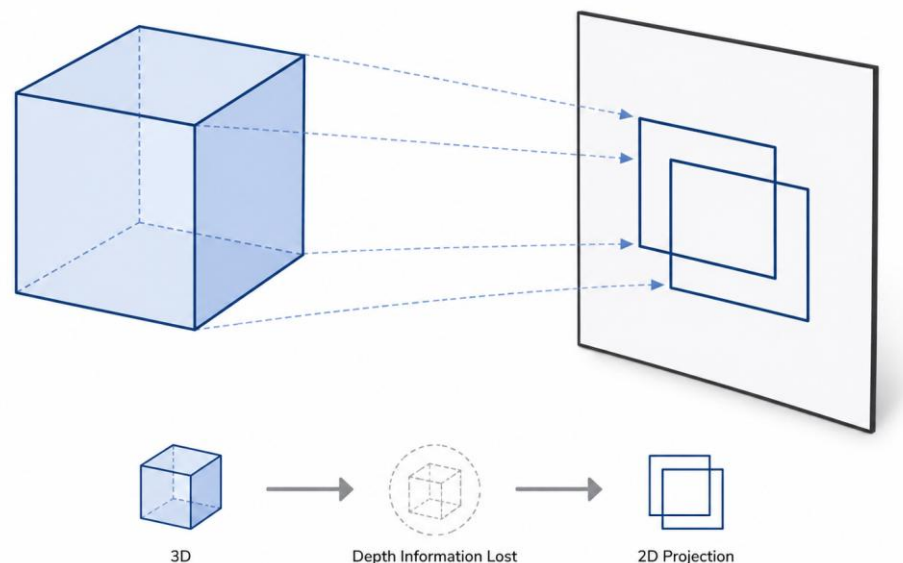
- Box plots summarize median, quartiles, spread, and outliers.
- Violin plots reveal distribution shape.





3D AND HIGHER-DIMENSIONAL DATA

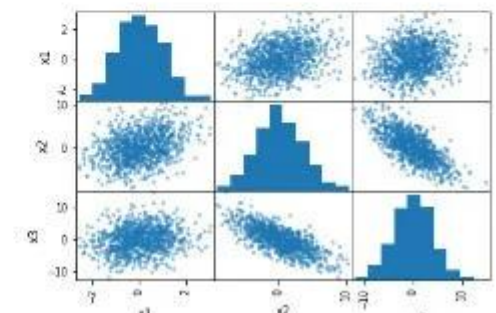
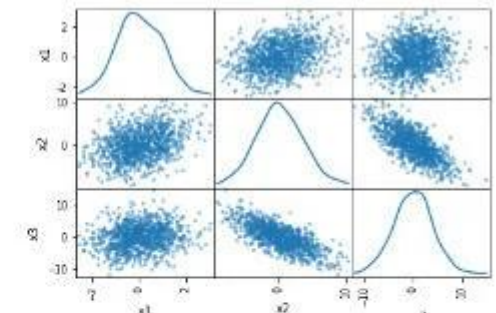
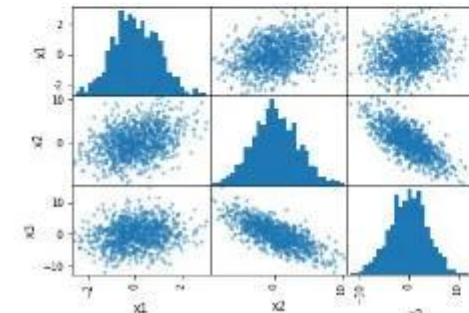
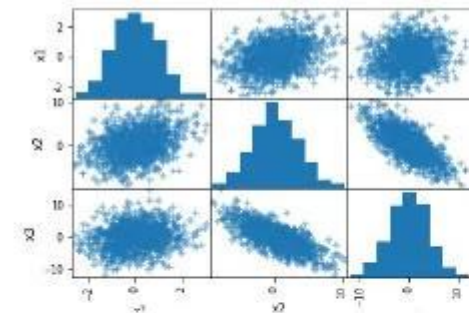
- 3D scatter plots can be hard to interpret on 2D screens.
- Prefer pairwise views, color, size, or interaction.





SCATTER MATRICES

- Scatter matrices show pairwise relationships across many variables.
- Great for exploration, too dense for presentation.

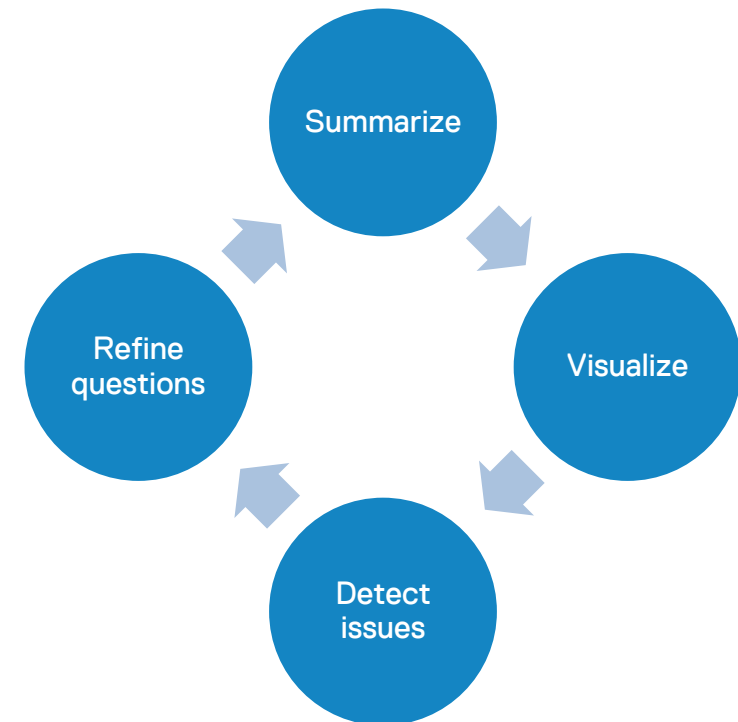




EXPLORATORY DATA ANALYSIS (EDA) TECHNIQUES

- Descriptive statistics
- Distribution analysis
- Outlier detection
- Correlation analysis

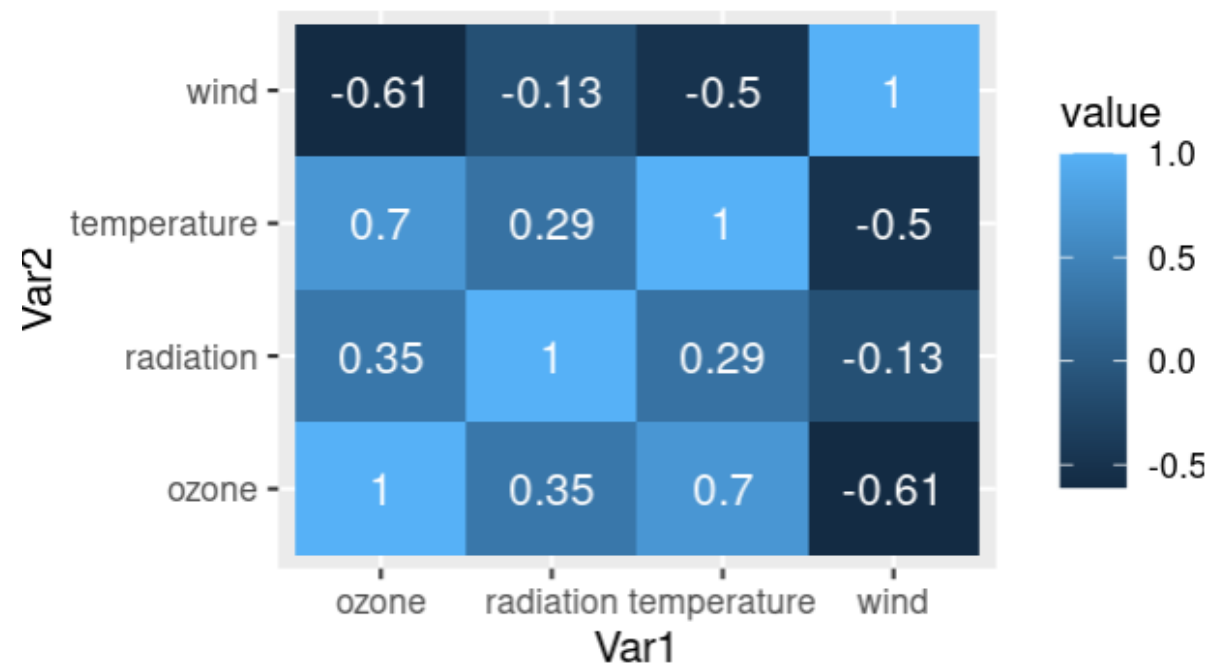
Guide preprocessing and modeling.





CORRELATION HEATMAPS

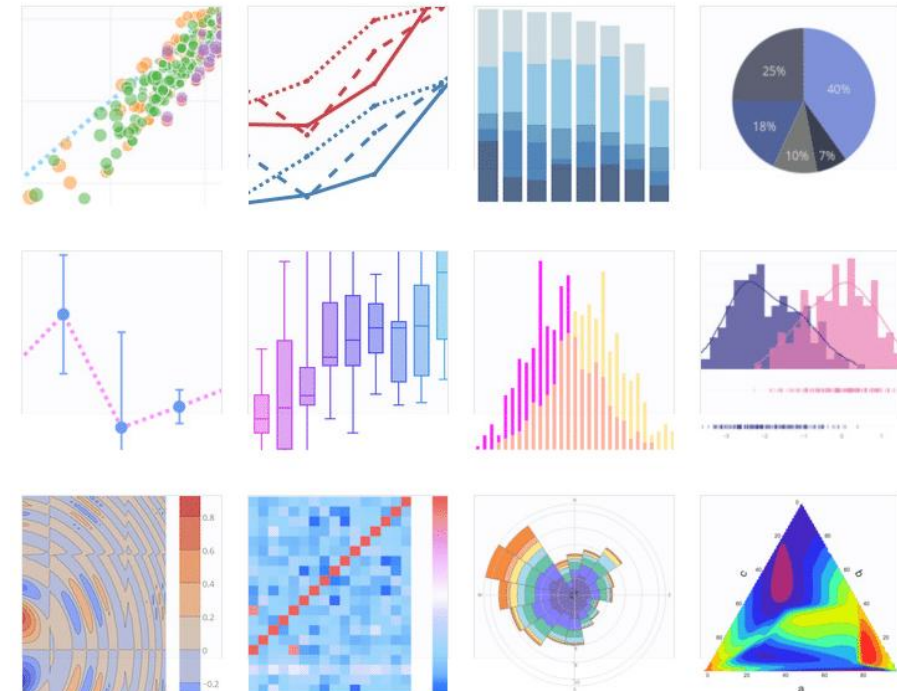
Correlation matrices help identify relationships, redundancy, and possible feature engineering ideas.





INTERACTIVE VISUALIZATIONS

- Interactivity supports zooming, hovering, filtering, and dashboard-based exploration.
- Plotly and Bokeh are useful options.





CHOOSING THE RIGHT CHART

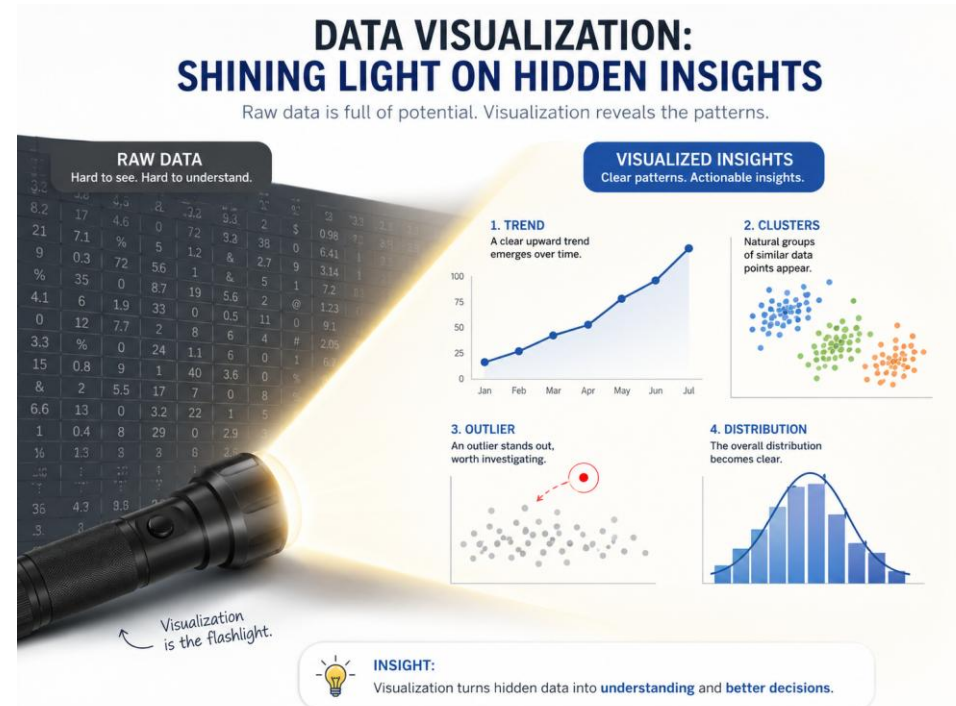
Match chart type to data type and question:

- ♦ Distribution
- ♦ Comparison
- ♦ Relationship
- ♦ Change over time
- ♦ Composition



KEY TAKEAWAYS

- Good visualization is not decoration; it is a thinking tool.
- Choose charts based on data type.
- Avoid misleading encodings and use interactivity when exploration benefits from it.



Technology Elevated